

LAB Logbook

GITHUB Link -

(All 10 labs)

Lab 1

DataFrame – The core Pandas class that represents tabular data in rows and columns, much like a spreadsheet or SQL table.

Series – A one-dimensional labeled array that can hold any data type, often used to represent a single column or row from a DataFrame.

Index – An immutable array used to label the axes of Series or DataFrames, enabling fast lookups and alignment.

Categorical – A class that represents a variable with a fixed number of possible values (categories), useful for memory-efficient encoding of repeated labels.

Timestamp – A class for representing individual points in time, offering powerful date-time functionality for time series analysis.

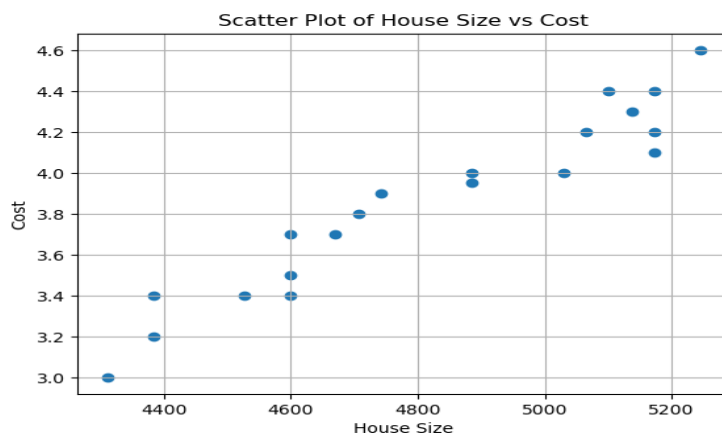
Lab 2

```
[45]: # Your code to create a scatter plot - size vs cost
import matplotlib.pyplot as plt

# Assuming house_size and cost are already defined from previous steps

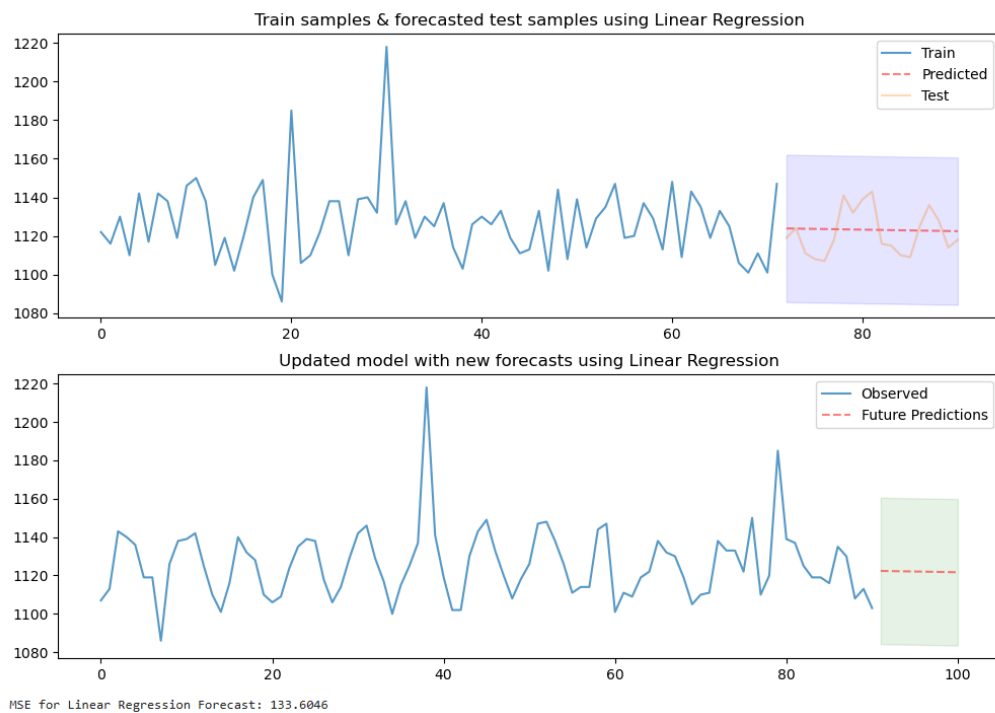
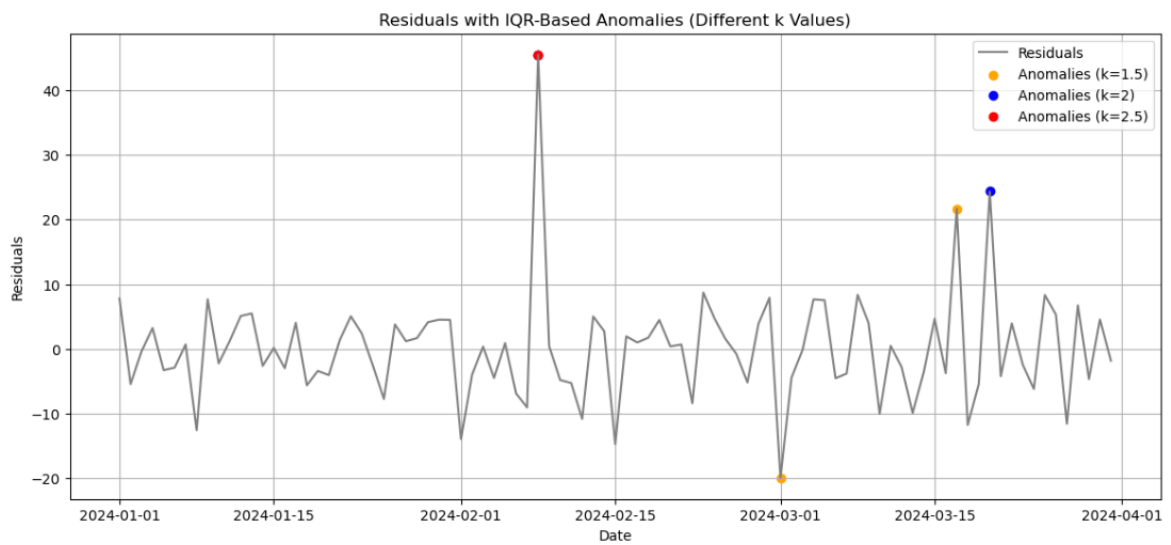
plt.scatter(house_size, cost)
plt.xlabel('House Size')
plt.ylabel('Cost')
plt.title('Scatter Plot of House Size vs Cost')
plt.grid(True)
plt.show()

size_guess = 2367185 * 0.75 # This equals 1775388.75
print(size_guess)
```



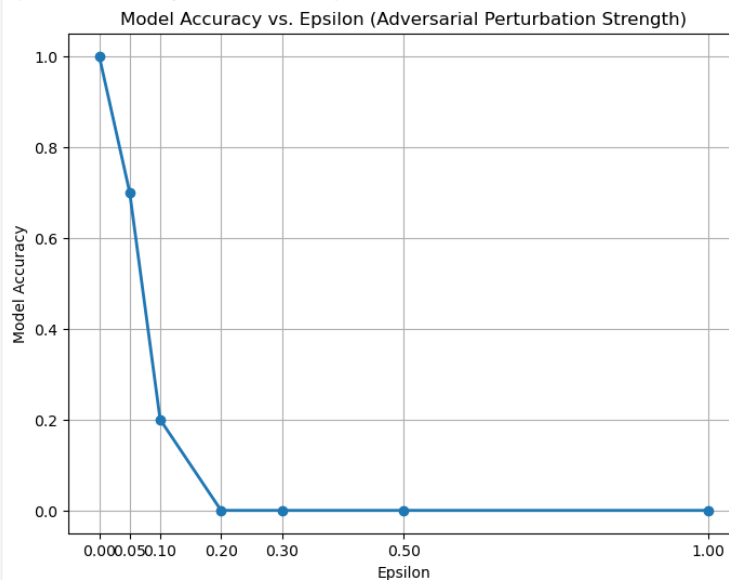
1775388.75

Lab 3



Lab 4

```
Epsilon: 0, Accuracy on adversarial examples: 1.0000
Epsilon: 0.05, Accuracy on adversarial examples: 0.7000
Epsilon: 0.1, Accuracy on adversarial examples: 0.2000
Epsilon: 0.2, Accuracy on adversarial examples: 0.0000
Epsilon: 0.3, Accuracy on adversarial examples: 0.0000
Epsilon: 0.5, Accuracy on adversarial examples: 0.0000
Epsilon: 1.0, Accuracy on adversarial examples: 0.0000
```



Lab 5

```
import time
from PIL import Image
import matplotlib.pyplot as plt
import tensorflow as tf

# Restore the last checkpoint (if available)
checkpoint.restore(tf.train.latest_checkpoint(checkpoint_dir))

# Set the required epoch number
epoch_no = 2

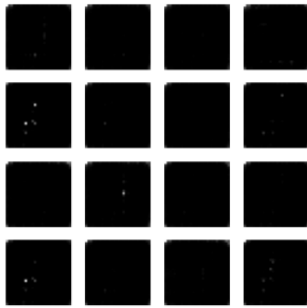
# Display the generated image for epoch 2
def display_image(epoch_no):
    return Image.open('image_at_epoch_{:04d}.png'.format(epoch_no))

# Call the function to display the image
img = display_image(epoch_no)

# Show the image using matplotlib
plt.imshow(img)
plt.axis('off')
plt.title(f'Generated Images at Epoch {epoch_no}')
plt.show()

# Print epoch number and simulated training time (replace with actual time if needed)
training_time = 30 # replace with the actual time in seconds for epoch 2
print(f"Epoch Number: {epoch_no}")
print(f"Training Time for Epoch {epoch_no}: {training_time} seconds")
```

Generated Images at Epoch 2



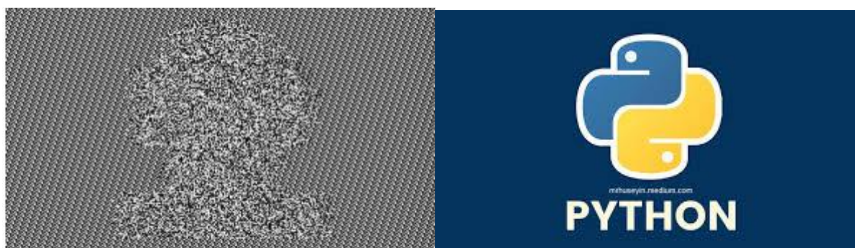
Epoch Number: 2
Training Time for Epoch 2: 30 seconds

```
[2]: generate_and_save_images(generator, 2, seed)
```



Lab 7

Input Secret Key 12345678
Enter Text python
Encrypted text: UNUhwmkCIqk=
Decrypted text: python



YES

Lab 8

Partner's name - Keerthana

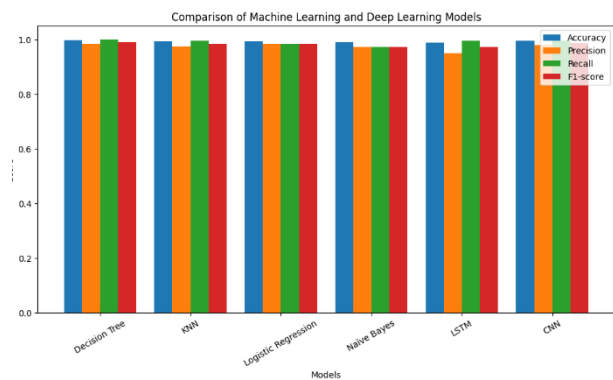
```
Public Values: p=21, g=7
Vishnu Private Key: 9
Vishnu Computed Shared Secret: 7
```

Lab 9

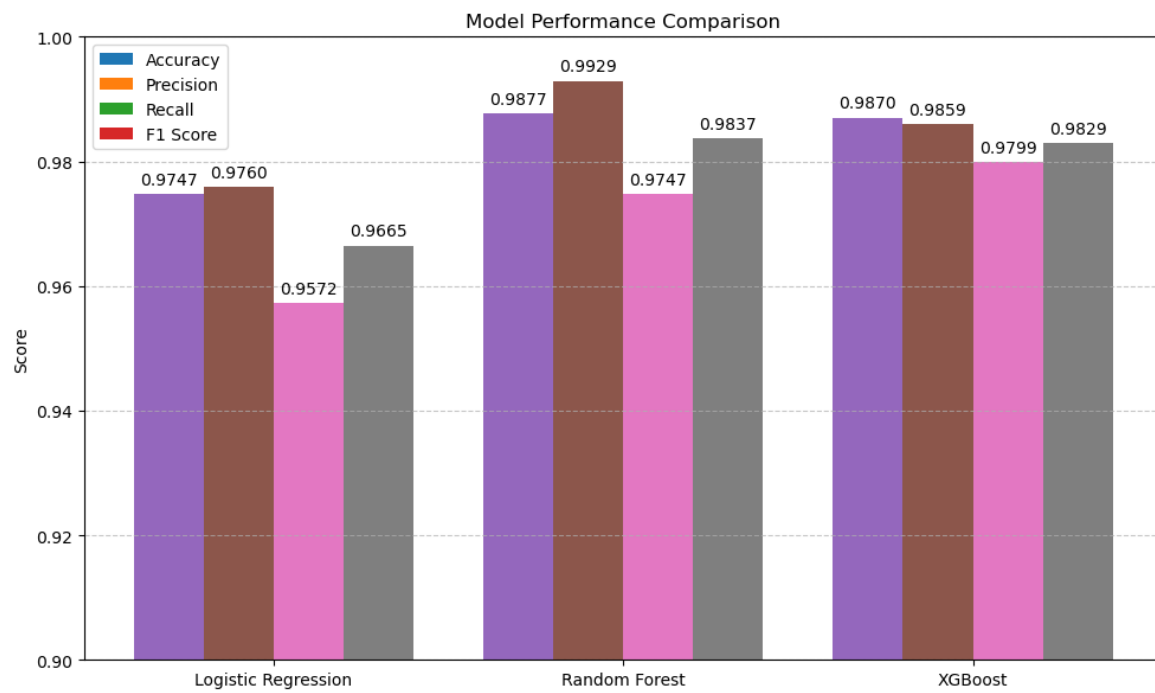
Lab Logbook Entry

- Attack Chosen: Twitter Bitcoin Scam (2020)
 - Useful Research Link:
[New York Times Report on Twitter Hack](#)
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Lab 10



Lab 11



Lab 12

No Activity for Lab Logbook (Exam)